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.xyz Presents

B^3

Build. Block. Break.

BASIC DETAILS

Project Name : ImageShield

Track Chosen: **Web3 for Good: Build What Matters**

Team Name: Misal Pav

Names of Members:
Sakshi,Samyuktha,Bhakti,Tanya
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ABOUT THE PROJECT

- Browser extension that embeds invisible watermarks into images before upload
- Watermark acts as a "GPS tracker" - detectable if image is stolen or misused
- Blockchain stores proof: wallet address + timestamp + image hash + tag ID
- Users get notified if their watermarked images appear in unauthorized places
- **Solves: No proof of ownership + no way to track stolen images + no legal evidence**

SCOPE OF PROJECT

- **Boundaries:**
  - We protect and track images at the point of upload via a browser extension.
  - We do not crawl the whole internet or scan all real AI training datasets yet.
- **Goals:**
  - Embed an invisible, robust watermark into user images before they are uploaded anywhere.
  - Anchor each watermarked image to a wallet address on-chain for provable ownership.
  - Provide a demo detector that can recognize our watermark and link it back to the owner.
- **Deliverables:**
  - Working browser extension that watermarks images on upload.
  - Deployed smart contract on Polygon testnet storing image/tag ownership.
  - Verification page to check if an image contains our watermark and show the on-chain record.

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image * required
string($binary)
Choose file download (2).jpg

tag_id * required
string
TEST_TAG_123

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/embed' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'image=@download (2).jpg;type=image/jpeg' \
  -F 'tag_id=TEST_TAG_123'
```

Request URL

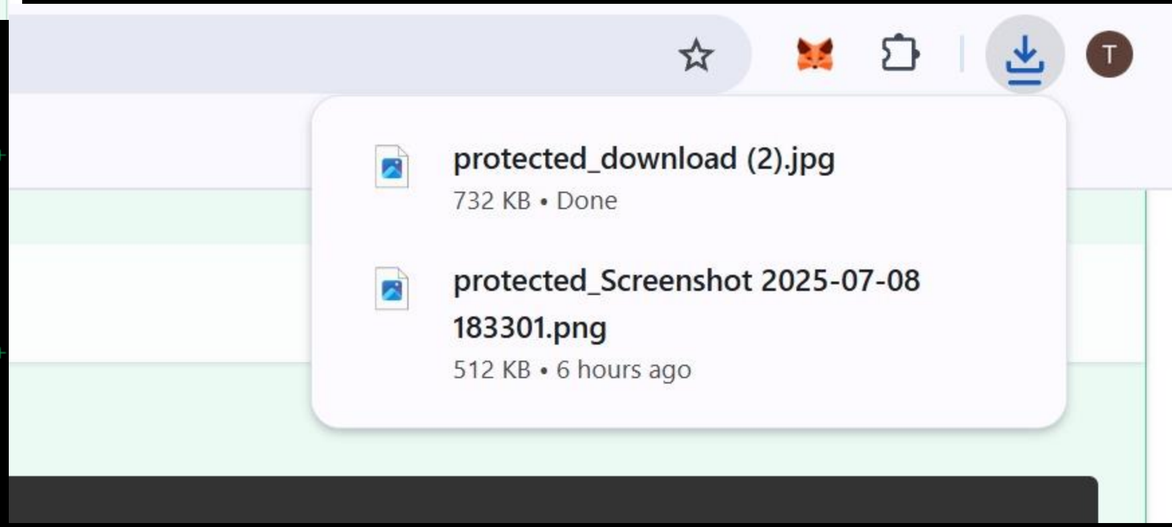
```
http://localhost:8000/embed
```

Server response

Code	Details
200	Response body Download file Response headers

Uploading image to generate water-marked image

Downloading the watermarked image



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MINGW64:/c/Users/Tanya/Documents/Projects/watermark-backend

```
Tanya@LAPTOP-LOVU41FG MINGW64 ~/Documents/Projects/watermark-backend (main)
$ curl -X POST -F "image=@test.jpg" -F "tag_id=GALLERY1" http://localhost:5000/a
pi/watermark-and-register -o gallery1.png
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  16372 100  16064 100    308  32561    624                              0

Tanya@LAPTOP-LOVU41FG MINGW64 ~/Documents/Projects/watermark-backend (main)
$ curl http://localhost:5000/api/gallery
{
  "images": [
    {
      "found": true,
      "image_hash": "abc123",
      "owner": "0x2030b953D0637C186E5F6Cb6F1E74E61424a5b74",
      "registered_date": "2026-02-08 09:39:24",
      "tag_id": "DIRECT1",
      "timestamp": 1770523764
    }
  ],
  "success": true,
  "total": 1
}
```

Proof that transparent full image watermark has been embedded in our image

.gitignore M | .env | deploy_contract.py U | app.py U | response_1770502998337.json | main.py | test999.png U

C: > Users > Tanya > Downloads > response_1770502998337.json > ...


```
{
  "success": true,
  "found": true,
  "tag_id": "\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0000\u0
  "message": "Watermark detected"
}
```

TIMELINE OF PROJECT

Till review 1: Invisible watermark UI
Till review 2: Blockchain integration
Till review 3: PPT and debugging
After review 3: Whole integration of backend and frontend, prototype ready

FUTURE SCOPE

Automated crawling to scan internet for stolen images and notify creators

Mobile app + partnerships with stock photo platforms and AI companies

Multi-chain support and NFT integration for tokenized ownership

Legal automation (auto-generate cease & desist) + enterprise licensing model

TECH STACK

Frontend: React, Chrome Extension API, Tailwind CSS

Watermarking: Python, Pillow, invisible-watermark library(Steganography)

Blockchain : Sepolia (Network in Ethereum) ethers.js

Backend: Flask , FastAPI ,Uvicorn

Web3: MetaMask wallet integration

Deployment: Railway, Vercel, Chrome Web Store